

Kalman Filtering-Aided Hybrid Indoor Positioning System with Fingerprinting and Multilateration

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Abstract—In this work we resort to the Bluetooth Low Energy (BLE) beaconing mechanism to propose an hybrid indoor positioning system (H-IPS) that fuses both multilateration (MLT) and fingerprinting (FP) RSSI-based approaches. The aim is to estimate the localization of an indoor target node, which is assumed to follow a uniform motion model. We adopt Kalman Filtering (KF) to diminish the MLT and FP errors while performing a track-to-track fusion (TTF) of the two KF outputs to further improve the performance. Our results indicate that the proposed H-IPS improves the estimation accuracy when individually compared to the standalone FP scheme in up to 46% in the considered scenarios, while the standalone MLT is outperformed in approximately 54%. Moreover, we also provide some insights on the influence of parameters such as the FP grid size, number of access points (APs) and number of samples on the accuracy of the proposed scheme. Finally, we show that the probability that the distance error of the proposed H-IPS is lower than 2 m is 92%, while for the FP and MLT the same probability is 43% and 47%, respectively.

Index Terms—Fingerprinting, Indoor Positioning System, Internet of Things, Kalman Filtering, Sensor Fusion, Multilateration.

I. INTRODUCTION

The Internet of Things (IoT) can be defined as a network of devices that are embedded in the environment, capable of sensing and/or influencing their surrounding without human interaction [1]. Location-based service (LBS) is an enabling feature for several IoT applications, such as indoor positioning (IP) [2]. However, traditional positioning techniques such as global positioning system (GPS) usually do not perform well in indoor environments, mainly due to shadowing effects [2]. Thus, alternative IP systems (IPS) have gained importance and attracted attention of the research community [3]–[11].

Some well-known localization techniques are based on the Received Signal Strength Indication (RSSI), Time of Arrival (ToA), Time Difference of Arrival (TDoA) and Angle of Arrival (AoA), among which RSSI-based techniques are usually adopted in the scope of Wireless Sensor Networks (WSN), since the other techniques often require additional hardware or advanced synchronization mechanisms [2].

RSSI-based localization approaches typically include a path-loss model to estimate, based on the received RSSIs, the distances from the access points (or anchor nodes) and then perform multilateration (MLT) [2], [12]. However, RSSI is affected by impairments inherent to radiofrequency (RF) trans-

missions, such as shadowing and multipath fading, besides being typically time varying, and subjected to interference from other devices operating at near frequencies [13].

The fingerprinting (FP) is a pattern matching technique capable of improving the performance of RSSI-based IPS [3]–[6], [8]. With FP, reference points are assigned to a collection of unique signal strengths, referred to as *fingerprints*, which are previously obtained in an offline fashion and stored in a database. FP is traditionally built indoors by means of WLAN (such as Wi-Fi) as the source of RSSI, *e.g.* [3]–[6]. However, it is shown in [7] that Bluetooth Low Energy (BLE)-based IPS provides a localization accuracy comparable to that of Wi-Fi, while presenting a much lower energy consumption and supporting portable battery-powered beacons that can be easily distributed at low cost. In [8], the authors show that BLE-based IPS are in fact capable of outperforming the accuracy of Wi-Fi FP-based IPS, depending on the density of the BLE network.

In scenarios where the target node (TN) is in movement and transmitting periodical beacons to the anchor nodes, one can improve the localization accuracy by combining the output of the RSSI-based IPS with the modeling pattern of the TN's movement. This can be done, for example, through Kalman Filters (KF) [14], which are an iterative mathematical process often applied in sensor fusion and position estimation [3], [9]–[11]. KF are widely used when the measured values contain unpredicted or random errors, and are based on linear minimum mean square error (MMSE) prediction and linear recurrent updating. Recursively, it considers as input both the previous predicted and the current measured values, outputting a more accurate prediction of the current position.

In [11], the authors propose a BLE-based IPS that resorts to FP and KF. More specifically, [11] adopts the K-nearest neighbor (KNN) technique with Bayesian estimation, where the signal fluctuations are also filtrated by KF, showing that accuracy can be increased in up to 40% when compared to standard KNN-based FP.

Aiming at combining the individual benefits of FP and MLT, in this work we propose an hybrid IPS (H-IPS) that fuses such schemes with the aid of KF and the track-to-track fusion (TTF) technique [15]. By considering that the TN follows a uniform motion model, we show through simulation results that the proposed H-IPS considerably improves the localization accuracy when compared to the standalone FP

and MLT techniques.

The remainder of the paper is organized as follows: Section II presents the system model and some preliminaries. The proposed KF-aided H-IPS is presented in Section III. Section IV presents numerical results, and Section V concludes the paper.

Notation: Throughout this paper, lowercase and uppercase boldface letters refer respectively to vectors and matrices, with $\mathbf{A}^T$ and $\mathbf{A}^{-1}$ being the transpose and the inverse of matrix $\mathbf{A}$, respectively. Moreover, $x \sim \mathcal{N}(a, b)$ represents a normal distributed random variable with average a and variance b , while $|A|$ represents the cardinality (number of elements) of a given set A .

II. SYSTEM MODEL AND PRELIMINARIES

We consider a rectangular Area of Interest (AoI) with dimensions $l \times w$ meters, where N stationary access points $\{\text{AP}_n\}_{n=1}^N$ are placed at known coordinates $(x_n, y_n) \in \text{AoI}$. The aim is to estimate the position of a Target Node (TN) that moves according to an uniform motion model. The TN periodically broadcasts a beacon signal, containing its Bluetooth Address (BD_ADDR), which is subjected to the effects of path-loss, shadowing and fading [13]. By considering that the effects of fast fading can be removed through averaging a large enough ensemble of independent samples [16], the RSSI (in dBm) at the n th AP can be estimated following a log-normal propagation model as [13]

$$\text{RSSI}_n = \text{RSSI}_{n,d_0} - 10\alpha \log_{10}(d_n/d_0) + X_\sigma, \quad (1)$$

where RSSI_{n,d_0} is the RSSI (in dBm) at AP_n when the TN is at a reference distance d_0 , d_n the distance between AP_n and TN, α the path loss exponent and $X_\sigma \sim \mathcal{N}(0, \sigma^2)$ represents the shadowing (in dB).

A. Multilateration (MLT)

Multilateration is a widespread technique used to estimate the TN location based on its distance measurements to $N \geq 3$ non-collinear APs. The distance between AP_n and the TN placed at coordinates (x, y) is calculated as $d_n = \sqrt{(x - x_n)^2 + (y - y_n)^2}$.

From (1), d_n can be estimated as $\hat{d}_n = d_0 10^{(\text{RSSI}_{n,d_0} - \text{RSSI}_n)/10\alpha}$. Assuming that $\hat{d}_n \simeq d_n$, squaring and subtracting the N th equation from the n th equation, and finally rearranging the terms, a linear function of x and y is obtained as [12]

$$\begin{aligned} -x_n^2 - y_n^2 + x_N^2 + y_N^2 + (\hat{d}_n)^2 - (\hat{d}_N)^2 = \\ x(-2x_n + 2x_N) + y(-2y_n + 2y_N). \end{aligned} \quad (2)$$

Representing the N equations as $\mathbf{b} = \mathbf{A}\mathbf{p}$ we have

$$\begin{bmatrix} -x_1^2 - y_1^2 + x_N^2 + y_N^2 + (\hat{d}_1)^2 - (\hat{d}_N)^2 \\ -x_2^2 - y_2^2 + x_N^2 + y_N^2 + (\hat{d}_2)^2 - (\hat{d}_N)^2 \\ \vdots \\ -x_{N-1}^2 - y_{N-1}^2 + x_N^2 + y_N^2 + (\hat{d}_{N-1})^2 - (\hat{d}_N)^2 \end{bmatrix} = \begin{bmatrix} -2x_1 + 2x_N & -2y_1 + 2y_N \\ -2x_2 + 2x_N & -2y_2 + 2y_N \\ \vdots & \vdots \\ -2x_{N-1} + 2x_N & -2y_{N-1} + 2y_N \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}. \quad (3)$$

Since $\{\hat{d}_n\}_{n=1}^N$ is subject to errors due to the shadowing effect, we obtain the least square estimate $\hat{\mathbf{p}}_{\text{MLT}} = \hat{\mathbf{p}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$ [12] so that we minimize $\sum_{n=1}^N (d_n - \hat{d}_n)^2$.

B. Fingerprinting (FP)

Fingerprinting is a pattern matching IPS where the signal received by the APs after a transmission performed by the tracked device is compared to a pre-defined database, composed of radio signatures obtained in an offline fashion from Fingerprint Reference Points (FRP) [5]. Fingerprinting involves two phases, namely the *calibration phase* (offline), and the *localization phase* (online) [3], [9], [17].

1) *Calibration phase:* The calibration phase is responsible for generating the fingerprinting database, which consists of the measures $\text{RSSI}_{m,n}$, i.e., the values of RSSI measured at $\{\text{AP}_n\}_{n=1}^N$, from a known device that sends beacons from a pre-established number of M equally spaced reference points $\{\text{FRP}_m\}_{m=1}^M$, positioned at coordinates $(x_m, y_m) \in \text{AoI}$.

Thus, the output of the calibration phase is composed of the identification (ID) of the AP_n and FRP_m , along with their locations (x_n, y_n) and (x_m, y_m) and corresponding $\text{RSSI}_{m,n}$ value, $\forall n \in \{1, \dots, N\}, \forall m \in \{1, \dots, M\}$ [3], [17]. Such database is assumed to be stored in a localization server connected to the APs, which will be responsible for estimating the position of TN in the online localization phase.

2) *Localization phase:* In this online phase, the TN broadcasts periodic beacons, whose RSSI is estimated by the APs and then forwarded to the localization server. The server is then responsible for comparing this set of RSSIs to the values stored in the database [3], [17]. As commonly adopted in the literature, in this paper we use a non-parametric data mining algorithm based on the Euclidean distance to calculate the distance between the set of RSSI values from the database ($\text{RSSI}_{m,n}$) and the current fingerprint measured by the device being tracked (RSSI_n), as follows [3]:

$$D_m = \sqrt{\sum_{n=1}^N (\text{RSSI}_n - \text{RSSI}_{m,n})^2}, \quad (4)$$

which leads to a set of distances $\mathcal{D} = \{D_1, \dots, D_m, \dots, D_M\}$, with $|\mathcal{D}| = M$. The estimated position of TN, which we refer to as $\hat{\mathbf{p}}_{\text{FP}} = (\hat{x}_{\text{FP}}, \hat{y}_{\text{FP}})$, is then obtained through the K-Nearest Neighbors (KNN) technique [11], whose output is the centroid of the polygon formed by the κ FRPs with the lowest D_m , that is, the first κ

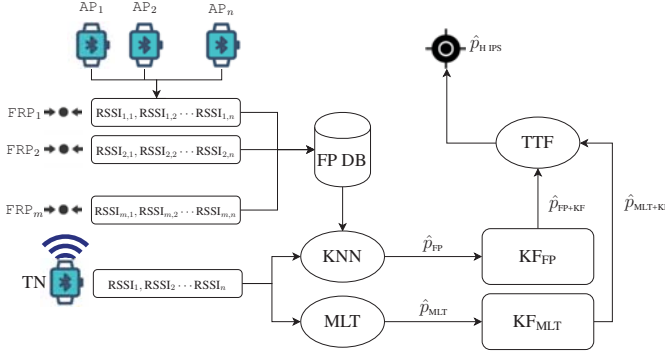


Fig. 1. Diagram model of the proposed KF-aided hybrid IPS.

elements of the set $\mathcal{D}_{\text{sort}} = \text{sort}(\mathcal{D})$, where $\text{sort}()$ corresponds to the sorting function in ascending order. In this work we adopt $\kappa = 4$, which is in accordance to values adopted in the literature [18] and has been found through simulations to establish a good trade-off between accuracy and complexity.

Due to the shadowing effect, the RSSI behaves as a random variable [19]. This leads to the selection of FRPs that are not necessarily the κ nearest neighbors, affecting the accuracy of RSSI-based methods, as is the case of fingerprinting and multilateration. Thus, in what follows we resort to Kalman filtering in order to mitigate the harmful effect of the randomness inherent to the wireless channel, while considering an uniform motion model.

III. KALMAN FILTERING-AIDED HYBRID IPS

A. Kalman Filter

We propose that the estimated positions $\hat{p}_{\text{MLT}}$ and $\hat{p}_{\text{FP}}$ outputted respectively by the multilateration and fingerprinting IPS are each an input to a different Kalman Filter (KF), along with an uniform motion model, aiming at smoothing the randomness inherent to the RSSI measures and consequently improving accuracy. To this end, we design two second-order filters with no external control inputs, which output a pair of 2D coordinates $\hat{p}_{j+\text{KF}} = (\hat{x}_{j+\text{KF}}, \hat{y}_{j+\text{KF}})$, $j \in \{\text{MLT}, \text{FP}\}$.

The uniform motion model considered in this work assumes that the TN moves with a constant velocity in both axis, *i.e.*, an object moving in a straight line with null acceleration. Thus, the state vector $\mathbf{x}_k$ is defined as the vector that contains the true values of the state variables at time k [9]:

$$\mathbf{x}_k = [x(k) \ y(k) \ v_x(k) \ v_y(k)]^T, \quad (5)$$

being $x(k), y(k), v_x(k), v_y(k)$ location coordinates and the velocities in x and y directions, respectively.

The measurement update function is calculated as [9]

$$\mathbf{z}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k, \quad (6)$$

where $\mathbf{H} \in \mathbb{R}^{2 \times 4}$ is the measurement model and $\mathbf{v}_k \in \mathbb{R}^{2 \times 1} \sim \mathcal{N}(0, \mathbf{R})$ represents the measurement noise, whose covariance matrix is $\mathbf{R} = r\mathbf{I}_2 \in \mathbb{R}^{2 \times 2}$, being $\mathbf{I}_i$ an $i \times i$ identity matrix and $r \in \mathbb{R}$. The Kalman filtering process can then be separated in two stages, namely the *time update* (which

is responsible for prediction) and the *measurement update* (correction) stages [9].

a) *Time update*: The state variables from (5) are updated according to the transition process [9]

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{A}\hat{\mathbf{x}}_{k-1|k-1}, \quad (7a)$$

$$\mathbf{P}_{k|k-1} = \mathbf{A}\mathbf{P}_{k-1|k-1}\mathbf{A}^T + \mathbf{Q}, \quad (7b)$$

where $\hat{\mathbf{x}}_{k|k-1} \in \mathbb{R}^{4 \times 1}$ is the a priori state estimate at time k given the a posteriori state estimate at time $k-1$, $\mathbf{A} \in \mathbb{R}^{4 \times 4}$ is the state transition model, $\mathbf{P}_{k|k-1} \in \mathbb{R}^{4 \times 4}$ is a priori error covariance matrix that collects the deviation in the estimations at time k given the a posteriori error covariance matrix at time $k-1$, $\mathbf{Q} = q\mathbf{I}_4 \in \mathbb{R}^{4 \times 4}$ is the covariance matrix of the transition model and $q \in \mathbb{R}$.

b) *Measurement update*: This stage combines the information from the current prediction to the information obtained from current observation in order to refine the state estimate [9]:

$$\mathbf{G}_k = \mathbf{P}_{k|k-1}\mathbf{H}^T(\mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^T + \mathbf{R})^{-1}, \quad (8a)$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{G}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1}), \quad (8b)$$

$$\mathbf{P}_{k|k} = (\mathbf{I}_4 - \mathbf{G}_k\mathbf{H})\mathbf{P}_{k|k-1}, \quad (8c)$$

being $\mathbf{G}_k \in \mathbb{R}^{4 \times 2}$ the Kalman Gain, which is responsible to adjust the trade-off between the prediction and the current measurement, $\hat{\mathbf{x}}_{k|k}$ the a posteriori state estimate given the measurements up to time k , $\mathbf{z}_k \in \mathbb{R}^{2 \times 1}$ the real measurement at time k and $\mathbf{P}_{k|k}$ the a posteriori error covariance matrix at time k .

B. Track-to-track fusion (TTF)

In order to extract the maximum amount of information possible about the TN location, in our H-IPS scheme we combine $\hat{p}_{\text{MLT}+\text{KF}}$ and $\hat{p}_{\text{FP}+\text{KF}}$ through the Track-to-track fusion technique. This is illustrated in Fig. 1. The TTF scheme was selected since it has lower computational cost when compared to measurement fusion, while presenting a better performance than modified TTF techniques when similar sensors are used, *e.g.*, two RSSI-based techniques [15].

The TTF is performed as [15]

$$\hat{\mathbf{x}}_{\text{H-IPS}} = \hat{\mathbf{x}}_{\text{MLT}+\text{KF}} + \frac{\mathbf{P}_{\text{MLT}}}{\mathbf{P}_{\text{MLT}} + \mathbf{P}_{\text{FP}}}(\hat{\mathbf{x}}_{\text{FP}+\text{KF}} - \hat{\mathbf{x}}_{\text{MLT}+\text{KF}}), \quad (9)$$

where $\hat{\mathbf{x}}_{j+\text{KF}}$ is the a posteriori state estimate for the $j \in \{\text{MLT}, \text{FP}\}$ technique, while $\mathbf{P}_j$ is the a posteriori error covariance matrix for the $j \in \{\text{MLT}, \text{FP}\}$ scheme.

IV. NUMERICAL RESULTS

In order to perform our evaluations, the received RSSI values were simulated through (1). The random variable representing X_σ is updated after every 100 RSSI measurements, for both FP and MLT algorithms. The RSSI of 1,000 beacons were averaged at each FRP during the calibration phase, as well as the localization phase, aiming at mitigating the effects of fast fading. The number of samples along the path was chosen as 200, which has been shown through simulation to

TABLE I
SYSTEM PARAMETERS

Parameter	Value	Parameter	Value
Path loss exponent (α)	1.8	RSSI measured at d_0 (RSSI_{n,d_0})	-52.36 dBm
Shadowing variance (σ^2)	4.57 dB	Reference distance (d_0)	1 m
x -axis velocity (v_x)	1.18 m/s	Room length (l)	60 m
y -axis velocity (v_y)	1.62 m/s	Room width (w)	40 m
Samples along the path	200		

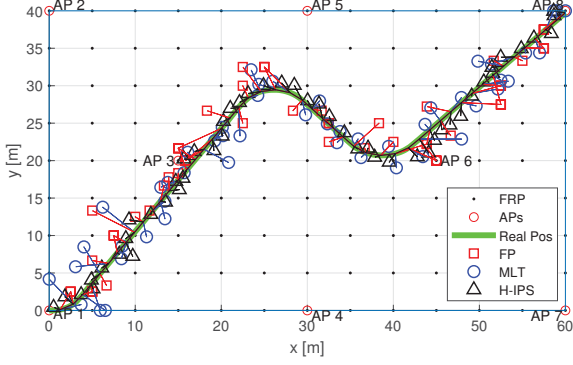


Fig. 2. Real Position, FP, MLT and H-IPS outputs for $l = 60$ m, $w = 40$ m, $N = 8$, FRPs' grid size is $5 \text{ m} \times 5 \text{ m}$ and 200 steps along the path, with markers at every 4 steps.

establish a good tradeoff between error floor and simulation complexity. Unless stated otherwise, the system parameters adopted in this section are the ones presented in Table I, which are in accordance to the literature [13], [20].

Fig. 2 considers a scenario with $N = 8$ APs and FRPs' grid size $5 \text{ m} \times 5 \text{ m}$, and shows the real path traveled by the TN (green boldface line) along with the corresponding location estimates provided by the MLT, FP and the proposed H-IPS techniques, the later with numerically obtained tuned values of r and q . It can be seen that the proposed H-IPS is capable of smoothing the errors, providing a much closer estimate to the real position.

The influence of the FRP grid size on the average error and on the database size is evaluated in Fig. 3, considering $N = 8$ APs and the following grid sizes: $10 \text{ m} \times 10 \text{ m}$, $8 \text{ m} \times 8 \text{ m}$, $5 \text{ m} \times 5 \text{ m}$, $3 \text{ m} \times 3 \text{ m}$ and $1 \text{ m} \times 1 \text{ m}$. Although the accuracy of the FP-based techniques increases when the grid size is reduced, the database size exponentially increases, which leads to expensive computations. For example, a reduction of the grid size from $10 \text{ m} \times 10 \text{ m}$ to $1 \text{ m} \times 1 \text{ m}$ decreases the error by 50.71% and a 57.98% for the FP and FP+KF, respectively. On the other hand, the database size increases from 280 RSSI measurements to 20,008, ~10,000% larger.

Nevertheless, it can also be seen in Fig. 3 that, for this number of APs, the accuracy of the proposed H-IPS scheme presents a small variation with the grid size, following the MLT-based schemes, which do not depend on the grid size.

Table II evaluates the influence of the number of APs in the average error of the IPS, for a grid size of $3 \text{ m} \times 3 \text{ m}$. In the aforementioned table, "FP+KF tuned" and "MLT+KF tuned" refer respectively to the FP+KF and MLT+KF schemes

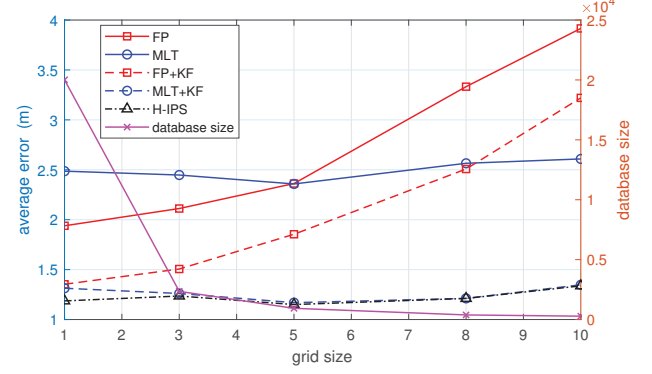


Fig. 3. Average error and database size versus the grid size, for $N = 8$ APs.

TABLE II
AVERAGE ERROR (IN METERS) FOR DIFFERENT NUMBER OF APs

# APs	3	4	5	6	7	8	9
FP	4.17	3.93	3.21	2.56	2.31	2.11	2.10
FP+KF	3.59	2.95	2.42	2.06	1.79	1.75	1.65
FP+KF tuned	2.97	1.84	1.70	1.63	1.69	1.51	1.26
MLT	5.25	4.32	3.46	3.20	2.69	2.45	2.50
MLT+KF	3.94	3.10	2.46	2.26	1.84	1.78	1.79
MLT+KF tuned	2.72	1.74	1.64	1.46	1.23	1.26	1.25
Proposed H-IPS	2.69	1.60	1.50	1.35	1.22	1.24	1.14

TABLE III
 r THAT MINIMIZES THE AVERAGE KF ERROR

# APs	3	4	5	6	7	8	9
$r_{\text{tun,FP}}$	40	70	40	30	20	30	20
$r_{\text{tun,MLT}}$	50	60	20	20	20	20	20

when adopting $q = 10$ and the tuned values of r , which are obtained numerically and provided in Table III. The results from the previously considered $r = 1$ and $q = 1$ scenario is also shown, and it can be seen that tuning r significantly improves the KF accuracy. Also, the proposed H-IPS scheme will have the best performance, regardless the number of APs.

Moreover, one can also see that the proposed H-IPS with only $N = 3$ APs can achieve a similar level of performance than that of the standalone MLT and FP schemes with a much higher $N = 9$ APs. This reduced number of APs necessary to achieve the same level of performance leads to a reduced database size and processing time. On the other hand, the accuracy of the proposed H-IPS with $N = 9$ APs outperforms the standalone MLT and FP techniques by approximately 54% and 46%, respectively. These values are highlighted in boldface letters in Table II.

In order to evaluate the KF transient, we show in Fig. 4 the variation of the error as a function of the number of samples utilized along the path. As expected, the FP and MLT have almost no performance variations with the number of samples. For the KF, in general, the error variance is larger in the initial steps, but it can be observed how this variance is reduced while increasing the number of steps due to the convergence of the

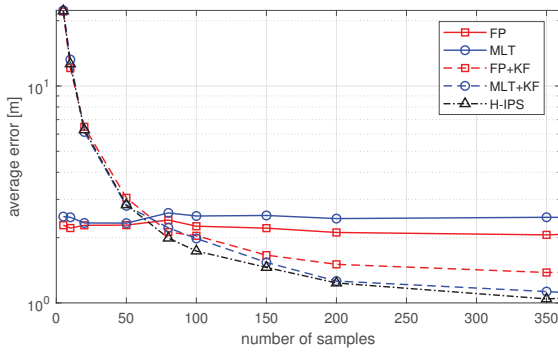


Fig. 4. Average error as a function of the number of samples. For ≥ 200 the error decrease rate is significantly reduced.

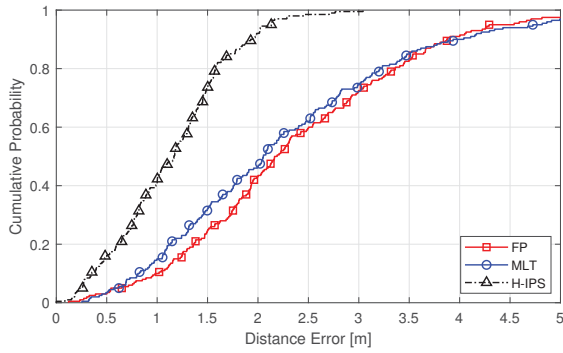


Fig. 5. Cumulative distribution function (CDF) of distance error for $N = 8$ APs and $5 \text{ m} \times 5 \text{ m}$ FRPs' grid size.

error covariance matrix. For ≥ 200 samples the error decrease rate is significantly reduced, so that we select the number of samples as 200 in our simulations.

Finally, in Fig. 5 we present the cumulative distribution function (CDF) of the distance error for the FP, MLT and the proposed H-IPS scheme, while considering $N = 8$ APs and $5 \text{ m} \times 5 \text{ m}$ FRPs' grid size. It can be seen that the probability that the error is lower than 2 m is 92% for our proposed scheme, while for the FP and MLT the same probability is 43% and 47%, respectively.

V. FINAL COMMENTS

In this work we proposed an hybrid indoor positioning system (H-IPS) that fuses both multilateration (MLT) and fingerprinting (FP) RSSI-based techniques, in a Bluetooth Low Energy (BLE) beaconing scenario. Moreover, we also adopted Kalman Filtering (KF) along with track-to-track fusion (TTF) in order to improve performance. Our results indicate that the proposed H-IPS improves the estimation accuracy when individually compared to the standalone FP scheme in up to 46%, while the standalone MLT is outperformed in approximately 54%. Also, insights on the influence of parameters such as the FP grid size, number of access points (APs) and number of samples on the accuracy of the proposed scheme are also provided. Finally, it is shown that the probability that

the distance error of the proposed scheme is lower than 2 m is 92%, while for the FP and MLT the same probability is 43% and 47%, respectively. As future work, the r and q tuning process can be performed dynamically, the KNN performance should be compared to the Support Vector Machine (SVM) and a practical implementation may validate the proposed scheme.

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